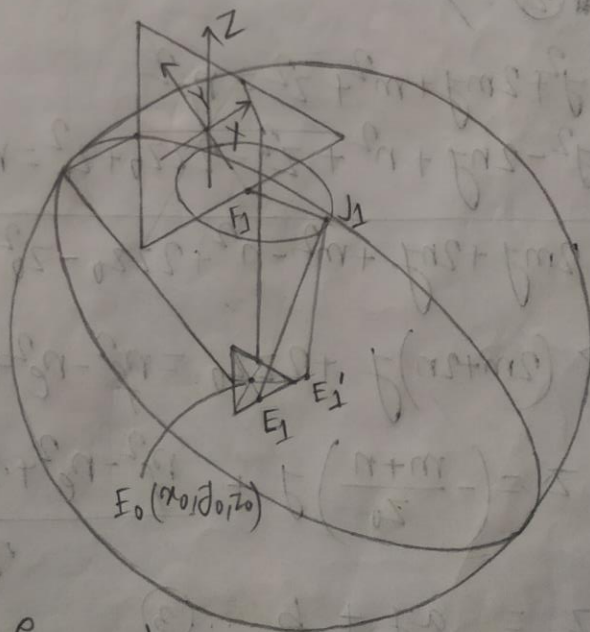
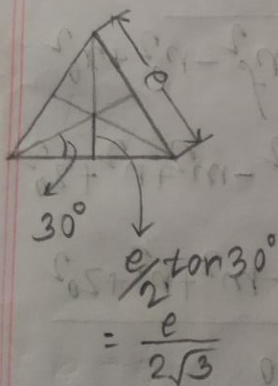
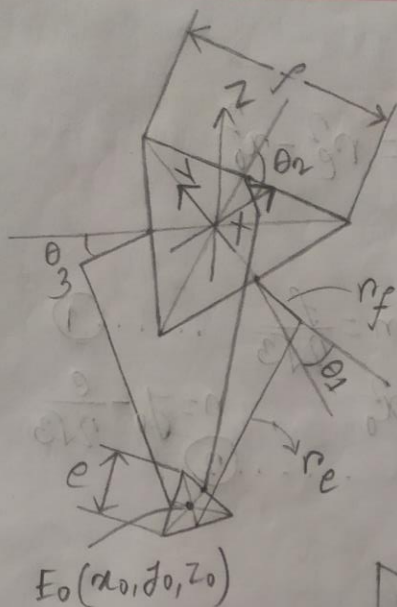


Inverse Kinematics of Delta Robot



$$\therefore E_1 \equiv (\alpha_0, j_0 - \frac{c}{2\sqrt{3}}, z_0)$$

$$E_A' \equiv (0, j_0 - \frac{c}{2\sqrt{3}}, z_0)$$

$$F_1 = \left(0, -\frac{f}{2\sqrt{3}}, 0\right), \quad J_1 = (0, f, z)$$

$$F_1 \sqrt{1} = r_p$$

$$E_1' \sqrt{1} = \sqrt{r_e^2 - \alpha_0^2}$$

$$\left(j + \frac{f}{2\sqrt{3}}\right)^2 + z^2 = r_f^2$$

$$\left(j - j_0 + \frac{e}{2\sqrt{3}}\right)^2 + (z - z_0)^2 = r_e^2 - \alpha_0^2$$

NOW

$$(j+m)^2 + z^2 = r_f^2, \quad m = \frac{f}{2\sqrt{3}} \dots\dots (1)$$

$$(j-n)^2 + (z-z_0)^2 = r_e^2 - \alpha_0^2, \quad n = j_0 - \frac{e}{2\sqrt{3}} \dots\dots (2)$$

$$(1) - (2) \Rightarrow$$

$$j^2 + 2mj + m^2 + z^2 = r_f^2$$

$$j^2 - 2nj + n^2 + z^2 - 2zz_0 + z_0^2 = r_e^2 - \alpha_0^2$$

$$\hline 2mj + 2nz + m^2 - n^2 + 2zz_0 - z_0^2 = r_f^2 - r_e^2 + \alpha_0^2$$

$$\Rightarrow (2m+2n)j + 2zz_0 = r_f^2 - r_e^2 + \alpha_0^2 - m^2 + n^2 + z_0^2$$

$$\Rightarrow z = \left(-\frac{m+n}{z_0}\right)j + \frac{r_f^2 - r_e^2 + \alpha_0^2 - m^2 + n^2 + z_0^2}{2z_0}$$

$$\therefore z = aj + b \dots (3)$$

$$a = -\frac{m+n}{z_0}, \quad b = \frac{r_f^2 - r_e^2 + \alpha_0^2 - m^2 + n^2 + z_0^2}{2z_0}$$

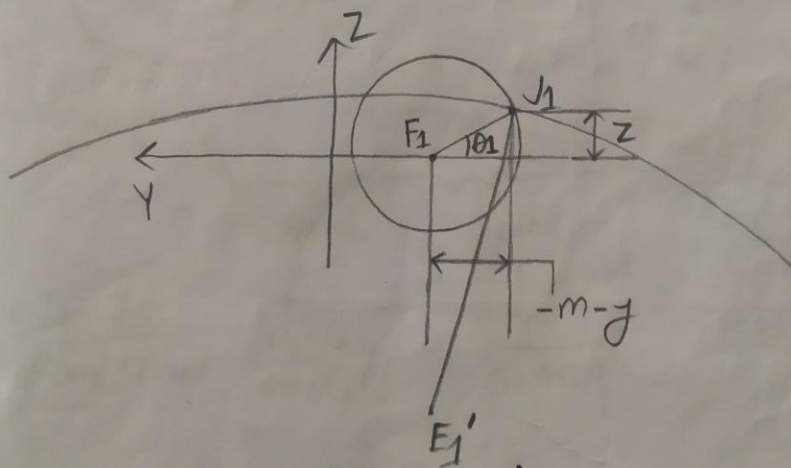
③ ① $\Rightarrow$

$$f^2 + 2mf + m^2 + a^2 f^2 + 2abf + b^2 = r_p^2$$

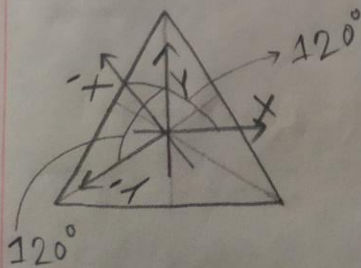
$$\Rightarrow (1+a^2)f^2 + (2m+2ab)f + m^2+b^2-r_p^2 = 0$$

$$\Rightarrow cf^2 + df + g = 0$$

$$\therefore f = \frac{-d \pm \sqrt{d^2 - 4cg}}{2c} \quad \therefore z = af + b$$



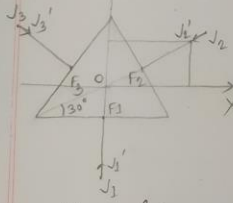
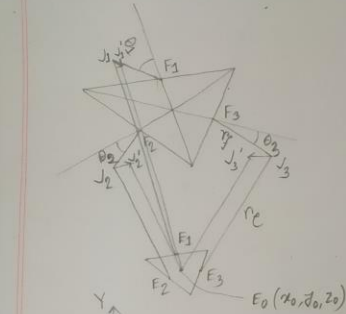
$$\theta_1 = -\tan^{-1}\left(\frac{z}{-m-f}\right)$$



$$X' = X\cos(120) + Y\sin(120)$$

$$Y' = -X\sin(120) + Y\cos(120)$$

Forward Kinematics of Delta Robot



$$OF_1 = OF_2 = OF_3 = \frac{f}{2 \tan 60^\circ} = \frac{f}{2\sqrt{3}}$$

$$J_1' - J_2' = J_3' - J_1' = \frac{e}{2 \tan 60^\circ} = \frac{e}{2\sqrt{3}}$$

$$F_1 J_1 = r_f \cos \theta_1 \quad F_2 J_2 = r_f \cos \theta_2$$

$$F_3 J_3 = r_f \cos \theta_3$$

$$J_1' \left(0, -\frac{f-e}{2\sqrt{3}}, -r_f \cos \theta_1, -r_f \sin \theta_1 \right) = J_1'(\alpha_1, d_1, z_1)$$

$$J_2' \left(\left[\frac{f-e}{2\sqrt{3}} + r_f \cos \theta_2 \right] \cos 30^\circ, \left[\frac{f-e}{2\sqrt{3}} + r_f \cos \theta_2 \right] \sin 30^\circ, -r_f \sin \theta_2 \right) = J_2'(\alpha_2, d_2, z_2)$$

$$J_3' \left(\left[\frac{f-e}{2\sqrt{3}} + r_f \cos \theta_3 \right] \cos 30^\circ, -\left[\frac{f-e}{2\sqrt{3}} + r_f \cos \theta_3 \right] \sin 30^\circ, -r_f \sin \theta_3 \right) = J_3'(\alpha_3, d_3, z_3)$$

Now

$$\alpha^2 + (d_1 - d_2)^2 + (z_1 - z_2)^2 = r_e^2 \dots (1)$$

$$(\alpha - \alpha_2)^2 + (d_1 - d_2)^2 + (z_1 - z_2)^2 = r_e^2 \dots (2)$$

$$(\alpha - \alpha_3)^2 + (d_1 - d_3)^2 + (z_1 - z_3)^2 = r_e^2 \dots (3)$$

Let

$$w_i = \frac{\alpha_i^2 + d_i^2 + z_i^2}{2}$$

$$\alpha_2 \alpha + (d_2 - d_1) d + (z_1 - z_2) z = \frac{w_1 - w_2}{2} \dots (4) = (1) - (2)$$

$$\alpha_3 \alpha + (d_1 - d_3) d + (z_1 - z_3) z = \frac{w_1 - w_3}{2} \dots (5) = (1) - (3)$$

$$(\alpha_2 - \alpha_3) \alpha + (d_2 - d_3) d + (z_2 - z_3) z = \frac{w_2 - w_3}{2} \dots (6) = (4) - (5)$$

$$(4), (5) \Rightarrow$$

$$\alpha = a_1 z + b_1 \dots (7)$$

$$d = a_2 z + b_2 \dots (8)$$

$$a_1 = \frac{1}{d} [(z_2 - z_1)(d_3 - d_1) - (z_3 - z_1)(d_2 - d_1)]$$

$$a_2 = -\frac{1}{d} [(z_2 - z_1) \alpha_3 - (z_3 - z_1) \alpha_2]$$

$$b_1 = -\frac{1}{2d} [(w_2 - w_1)(d_3 - d_1) - (w_3 - w_1)(d_2 - d_1)]$$

$$b_2 = \frac{1}{2d} [(w_2 - w_1) \alpha_3 - (w_3 - w_1) \alpha_2]$$

$$d = (d_2 - d_1) \alpha_3 - (d_3 - d_1) \alpha_2$$

$$(7), (8), (1) \Rightarrow$$

$$(\alpha_1^2 + \alpha_2^2 + 1) z^2 + 2[a_1 b_1 + a_2(b_2 - d_1) - z_1] z + [b_1^2 + (b_2 - d_1)^2 + z_1^2 - r_e^2] = 0$$